# Oriented Infrared Vehicle Detection in Aerial Images via Mining Frequency and Semantic Information

Nan Zhang, Youmeng Liu, Hao Liu, Tian Tian, and Jinwen Tian

***Abstract*— Infrared vehicle detection based on aerial images has significant applications in military and civilian fields for the perception ability under low-light and foggy conditions. However, it remains challenging due to the following characteristics. First, infrared textures and edges are blurred, implying a plenty of low-frequency signals and a shortage of detailed descriptions. Second, objects in infrared images present different patterns depending on their thermal radiations, which hamper the feature extraction of convolution kernels. Third, infrared images lack color information, which means that fewer features for classification and regression can be used. Inspired by cognitive neuroscience that humans perceive the entirety from low-frequency information and discern details from high-frequency information, we devise a new framework for oriented infrared vehicle detection called infrared information mining detector ($I^2$MDet) to tackle the above challenges. It consists of two significant designs: the kaleidoscope module and the semantic feature supplement module (SFSM). In the kaleidoscope module, we explore the effect of kernel sizes and dilation rates on frequency information mining with kaleidoscope-like equivalent kernels. Features in this module are extracted by adaptive involution operators instead of convolution kernels to deal with multiple patterns. The SFSM provides the network with features beneficial for classification and regression. On the one hand, the network is guided to learn more meaningful features under semantic supervision. On the other hand, features output by the SFSM supplement the detection head with semantic information. Experimental results on the public dataset DroneVehicle demonstrate that our proposed approach achieves outstanding performance on oriented infrared vehicle detection.**

***Index Terms*— Frequency features, infrared aerial image, oriented bounding box, semantic features, vehicle detection.**

## I. Introduction

VEHICLE detection is a major and challenging topic in remote sensing research. It plays a vital role in many applications, such as military target reconnaissance [1], traffic flow monitoring [2], [3], [4], [5], [6], [7], [8], disaster rescue, and security [9], [10], which has been attracting significant attention of researchers. Current research on vehicle detection mainly focuses on visible images. However, many detection tasks need to be carried out under low-light conditions or in bad weathers, such as reconnoitering military vehicles at night or monitoring traffic flow in foggy weather. Nevertheless, visible sensors cannot work well in low illumination and low visibility. In contrast, infrared sensors can handle full-time work even in some severe weather, and thus, it is motivated to research vehicle detection on infrared aerial images.

The prosperity of the generic object detection [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22] and oriented bounding box object detection [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42] has greatly promoted vehicle detection. Targets are usually represented by horizontal bounding boxes in generic object detection, which can satisfy the demand for detection in most scenarios. The oriented object detection surrounds targets by arbitrary-oriented quadrilaterals, which tackle the difficulty of targets with dense arrangement and arbitrary directions. However, these current approaches are proposed for visible images. They do not achieve satisfactory performance when applied to infrared images because infrared images have the following properties compared to visible images. First, we analyze from the perspective of signal frequency. The paired visible and infrared images are decomposed into several identical frequencies, as shown in Fig. 1. The blue and the red dotted boxes display the high- and low-frequency parts of the image, respectively. It is observed from the green circles of Fig. 1 that the signal of the infrared image primarily concentrates on the low-frequency but rarely appears in the high frequency. In the blue dotted box, the first column of the infrared image processed by a high-pass filter with a lower cutoff frequency contains more visible edges than the second column image. In the red dotted box, the second column of the infrared image processed by a low-pass filter with a higher cutoff frequency contains clearer shapes than the first column image. Therefore, we expect the low-pass filter to have a higher cutoff frequency and the high-pass filter to have a lower cutoff frequency in the network because a low-pass filter with a high cutoff frequency increases the high-frequency information and a high-pass filter with a low cutoff frequency increases the signal-to-noise ratio of the features.

Then, we consider it from the perspective of the imaging mechanism. The infrared image reflects the thermal radiation properties of objects. The objects show different patterns depending on the temperatures. However, the convolution kernel is a feature description operator applied to extract

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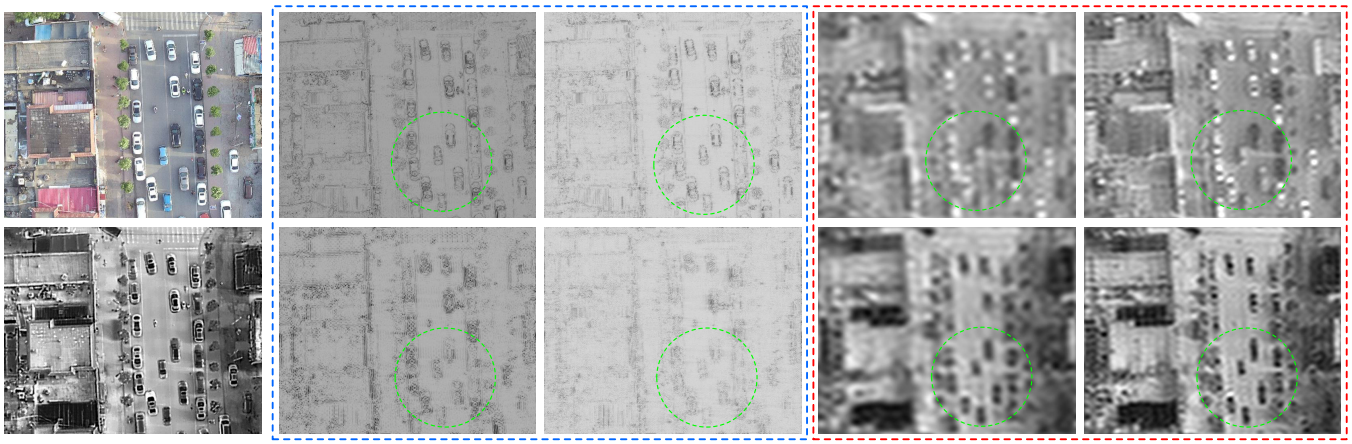


Fig. 1. Frequency decomposition illustration of a pair of infrared and visible images. The first row is the visible image and the second is the infrared image, where the two corresponding images above and below present the information of the same frequency. The two pairs of images in the blue boxes are the high-frequency parts. The first and second columns in the blue box are the results of high-pass filtering with filters of radius 50 and 100, respectively. The red box shows the low-frequency component of the image. The first and second columns in the red box are the results of low-pass filtering with filters of radius 20 and 30, respectively.

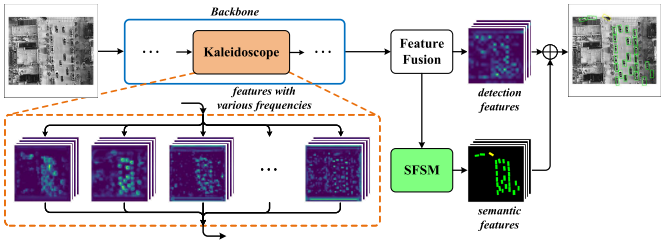



Fig. 2. Simplified structure of our I$^2$MDet. The orange dashed box displays the kaleidoscope module, which can extract various frequency features from the infrared image. The green module is the SFSM, which supplements the detection head with semantic features that contain information about the vehicle category, position, outline, and angle.

local invariant features. When the thermal radiation of the vehicle changes due to the movement state, the local features are changed. The original convolution kernel is not good at extracting this feature, which increases the learning difficulty. Finally, we analyze the amount of information. The infrared image is a single-channel grayscale image with no color information, which prevents the network to learn more semantic features compared to visible images. All these problems result in the inferior performance of infrared object detection when employing current deep learning detection networks.

To address the challenges mentioned above, we devise a novel framework for oriented infrared vehicle detection called infrared information mining detector (I$^2$MDet). The simplified overall framework is shown in Fig. 2. Our I$^2$MDet includes the kaleidoscope module to sufficiently extract diverse frequency information and a semantic feature supplement module (SFSM) to provide more information for the network. Sequentially, we will describe how these two improvements alleviate the challenges of infrared object detection. There are low-pass filters to extract the entirety and high-pass filters to extract the edges in the convolutional neural network. Based on the analysis of Fig. 1, we expect that: 1) the low-pass filter has a higher cutoff frequency, which will increase high-frequency components and diversity of features, and 2) the high-pass filter has a lower cutoff frequency, which will reduce the disturbance caused by noise in high-frequency components. We found that expanding the size of the kernel can increase the cutoff frequency of the low-pass filter, which will be proved in Section III-B. Increasing the dilation rate for the high-pass filter helps to increase the gradients when extracting the edge features, equivalent to reducing its cutoff frequency.

In addition, global self-attention proves to be better at extracting low-frequency signals [43], which makes full use of the low-frequency information in infrared images. We prefer the network to simultaneously hold these salient features extracted by different operators. Therefore, in the kaleidoscope, we use the Inception [44] structure to introduce the self-attention [45] and the involution [46] with different kernel sizes and dilation rates to focus on different frequencies of the image. Subsequently, the weight factors of different frequency features are provided by channel attention [47] to fuse the features of each frequency. The reason we employ the involution operator instead of the general convolution kernel to extract features in the kaleidoscope is that the involution operator is input-adaptive. It generates different weights to extract features according to the different patterns of objects, which improves the robustness of multiple patterns. As for the problem of insufficient features of classification and regression for infrared images, we propose the SFSM without semantic annotation costs to supplement the network with semantic information. It is added at the end of the detector. On the one hand, the SFSM shares the backbone and feature fusion network with the detection head, which guides the network to learn more meaningful features via semantic supervision during training. On the other hand, we supplement the features extracted from the SFSM to the detection head. The semantic features contain information about the category, location, outline, and angle of the object, which contributes to classification and regression. Due to the near-rectangular shape of the vehicle, the mask for semantic supervision can be generated directly from the oriented bounding boxes, which requires no additional annotations of the training data.

In summary, the contributions of this article are summarized as follows.

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1) We explore the relationship between kernel sizes and dilation rates on the frequency of extracted features, which is the principle for designing the kaleidoscope module.
2) The kaleidoscope module is devised to sufficiently exploit and integrate features in diverse frequencies of the image. Meanwhile, the input-adaptive involution in the kaleidoscope relieves the problem of infrared object pattern diversity.
3) We propose the SFSM to tackle the problem of insufficient features in infrared images. The SFSM guides the network to learn more meaningful features via semantic supervision and supplements the semantic features to the detection head.

The rest of this article is organized as follows. Section II summarizes some relevant works about oriented bounding box object detection, infrared vehicle detection, and introduction of information to the network. Section III describes the structure of our I$^2$MDet in detail. The results and discussions of experiments are demonstrated in Section IV. Some conclusions of this article are given in Section V.

## II. Related Work

### A. Oriented Bounding Box Object Detection

The approach of rotation detection is proposed to detect scene text [48], [49], [50], [51], [52], retail items [53], [54], and aerial images [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42]. Many approaches of oriented object detection are modified from the superior horizontal object detection [15], [16], [17], [18], [19], which directly regress an angle parameter. However, objects in aerial images have the characteristics of dense arrangement and arbitrary orientation, which suffer from feature coupling and blurred bounding. Therefore, the improvement is roughly divided into two aspects. On the one hand, how to get more aligned features of the object is investigated. Some two-stage methods [23], [26], [33] refine the region proposal network (RPN) to generate rotated proposal boxes and the ROI pooling to align rotated features. Han et al. [30] captured the features of oriented object by deformable convolution [55]. Hou et al. [35] assigned different samples selection strategies to targets with different aspect ratios to ensure the accuracy of the features. On the other hand, how to represent the object that is more beneficial for optimization is studied. The oriented object is initially represented by the horizontal rectangular box with an angle [48]. The angular periodicity causes a drastic change of the loss at the boundary. To address the issue of discontinuous boundaries, Yang and Yan [27] converted angular prediction from regression to classification. An intersection over union (IoU)-related factor is added to the loss to relieve this problem [24], [25], [26]. Yu and Da [56] mapped rotational periodicities of different cycles into phases of different frequencies, which provides a solution to the problem of the periodicity of the box angle. Then, some other representations of oriented targets are explored. The rotated bounding box is represented as a 2-D Gaussian distribution in the literature [36], [37], [38], [39]. They optimize the network by measuring the distance between two Gaussian distributions representing the prediction and the ground truth. The object is represented by a set of points to reflect its position more accurately [32], [34]. Xu et al. [28] and Qian et al. [29] predicted a horizontal bounding box and the offset of each vertex. The target is represented by a quadrilateral inside the horizontal bounding box. The object is described as a center point, four polar angles, and four polar diameters in the literature [41], [42]. The large-scale detection model for remote sensing is also explored [57]. However, all these approaches are proposed for visible image object detection. They adopt generic convolution kernels that are fixed after training, and thus, many features will be ignored due to the multiple patterns of infrared objects. The involution operator in our kaleidoscope module is input-adaptive, which learns the essential features behind the pattern diversity. Moreover, we improve the performance of oriented object detection by introducing the semantic information. The semantic label describes the category, location, and angle of the object. It not only reduces the complexity of optimization but also complements the detection head with features that are beneficial for classification and regression.

### B. Infrared Vehicle Detection

The challenges of infrared vehicle detection are the low-frequency features and the small amount of information. Chen et al. [58] enhanced the contrast between the vehicle and the background by morphology cap transformation to sharpen the outline. Du et al. [59] applied transfer learning from the visible dataset to the infrared dataset and mined negative samples due to insufficient infrared data. For infrared dim-small target detection, the motion features are introduced [60], [61], [62]. They filter false alarms by the characteristics of the object between multiple frames. Multimodal data fusion is also a standard method to increase features. The infrared image is fused with the RGB image in the literature [63], [64]. They combined the extensive spectral and textural features of visible images with infrared images to enhance the performance of the network. Yang et al. [5] fused light detection and ranging (LiDAR) point cloud cluster into infrared object detection to provide additional position and size information of the object. However, they require additional data to provide information for the infrared images, which is labor-intensive. The SFSM we design does not require additional data collection and manual annotation to train. The mask for semantic supervision is freely generated by the oriented bounding box. Our SFSM can introduce not only the object position and size but also angles and outlines. In addition, the kaleidoscope we proposed increases the feature frequency of not only the outline but also the whole map.

### C. Introducing Information Into the Network

The neural network is a data-driven approach, so there is no doubt that more information leads to better performance. Introducing information into the network can be roughly divided into two ways, introducing multiple types of data from the input side and introducing supervision by multiple loss

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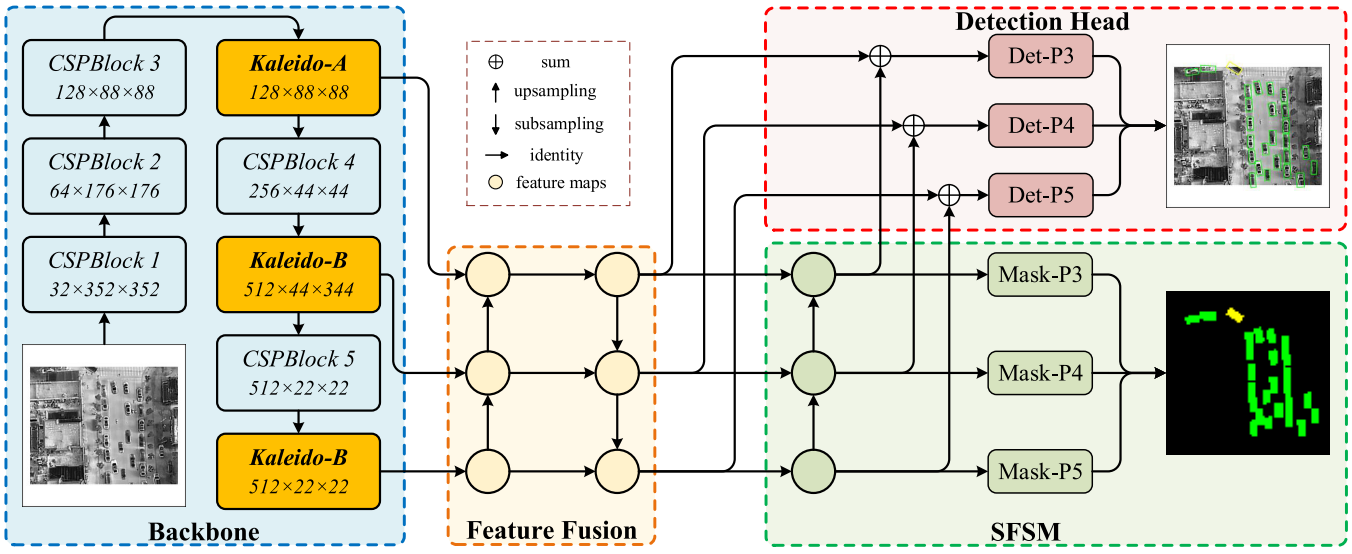



Fig. 3. Framework of the proposed I$^2$MDet for infrared image detection. The blue area is the backbone, which consists of kaleidoscope modules and CSPNet with dilation. We extract low-level features by Kaleido-A and high-level features by Kaleido-B. Kaleido-A does not contain the self-attention due to memory restrictions. The beige part is the feature fusion network, which adopts the top-down and bottom-up structure in PANet. The red area is the detection head, where Det is the decoder for predicting the oriented bounding box of the vehicle. The green part is the SFSM we proposed. It shares the backbone and feature fusion network with the detection head. The features it outputs, on the one hand, are supplemented to the detection head and, on the other hand, are used for supervision after the mask block.

functions from the output side. Multiple types of data used are also known as multimodal inputs. Except for the modes of the multimodal input introduced in Section II-B, Zhang et al. [65] integrated RGB images in the detection of point cloud vehicles. Tang et al. [66] improved the fusion effect with RGB images, infrared images, and semantic information. The digital surface model (DSM) is used as an auxiliary input to improve the accuracy of the mask boundaries in the semantic segmentation task [67]. The optimization of networks guided by multiple loss functions is commonly termed multitask learning. Many researchers adopt multitask learning to improve the performance of one of them because the network could learn new meaningful features supervised by associated tasks. For the semantic segmentation task, Yu et al. [68] employed two labels, object boundary masks and semantic masks, to guide network learning, described as decoupling the edge and texture features. Bertasius et al. [69] also introduced boundary supervision, but the difference is that the predicted boundary field is used to constrain the semantic mask during inference. SFM-Net predicts depth, segmentation, camera, and rigid object motions, converts those into point cloud data, and maps the 3-D point cloud to a 2-D motion field (optical flow) [70]. ResUNet [71] utilizes the distance transform of the segmentation mask and a colored reconstruction of the input as supervision except for the target boundary. As the detection task, Chen et al. [72] introduced the masks of foreground and target corner points as supervision. The foreground and corner points features are complementary to the regression features. The detection boxes can also be corrected by the predicted foreground and target corner points during inference. SCRDeT++ [25] and AFCN [73] both introduce semantic-level labels in the detection task. SCRDeT++ performs the instance-level denoising (InLD) for the feature maps by the predicted semantic mask. AFCN trains a separate semantic segmentation model and filters detector proposal boxes by the predicted mask. It is worth noting that SCRDeT++ and AFCN are similar to our approach because they both leverage semantic information in the detection task. However, there is an inherent difference in operation. We train the SFSM in I$^2$MDet at the end of the detector and complement the detection features with semantic features. Unlike AFCN, our network is guided to learn more significant features because of the shared backbone. Moreover, we supplement the detection features rather than filtering them by semantic features to avoid suppressing correct objects compared to SCRDeT++.

## III. Methods

### A. Overall Framework

The architecture of the detailed overall framework is shown in Fig. 3. It could be roughly divided into four parts.

1) The blue area is the backbone, which consists of kaleidoscope modules and CSPNet [74]. The CSPNet is a lightweight structure with excellent feature extraction performance. The kaleidoscope module is interspersed in CSPNet to extract features of various frequencies. There are two forms of the kaleidoscope module in the backbone, Kaleido-A for extracting low-level features and Kaleido-B for high-level features. The difference is that Kaleido-A does not contain self-attention since it consumes many resources to use self-attention in the underlying high-resolution feature maps.
2) The beige part is the feature fusion network, which fuses the three scale features output from the backbone. We adopt the top-down and bottom-up architectures in PANet [75], which can fuse low-level features with high frequency to enrich the details of high-level features.

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3) The red part is the detection head, which accepts the features from the feature fusion network and the SFSM and then outputs the detection results. Det decodes the features into parameters that represent the oriented bounding box.
4) The green part is the SFSM, which shares the backbone and feature fusion network with the detection head. The features it outputs are primarily used for two purposes: one is supplemented into the detection head and the other flows into the mask block. The mask block adjusts the semantic features to the one-hot encoding for supervision, which guides the network to learn more meaningful features. Specifically, we will describe the kaleidoscope and the SFSM in Sections III-B and III-C, respectively.

### B. Kaleidoscope

According to the convolution theorem, the convolution of the spatial domain is equal to the multiplication of the frequency domain. The formula for the convolution theorem is shown as follows:

$$\begin{aligned}\mathscr{F}[\text{feature map}] &= \mathscr{F}[f(x,y) * g(x,y)] \\ &= \mathscr{F}[f(x,y)] \times \mathscr{F}[g(x,y)] \end{aligned} \tag{1}$$

where $\mathscr{F}$ represents the Fourier transform, $f(x, y)$ represents the image signal, and $g(x, y)$ represents the filter signal. The signals are discrete in convolutional neural networks, so the image and filter signals must be sampled. The imaging device samples the image signal before it is fed into the network, which indicates that $f(x, y)$ is fixed. The filter signal is learned by the network and the filter size corresponds to the number of sampling. The kernel is the discrete filter signal. According to the sampling theorem

$$f_s \geq 2 f_{\max} \tag{2}$$

where $f_s$ is the frequency of sampling and $f_{\max}$ is the maximum frequency of the signal, and the sampling frequency is two times greater than the maximum frequency of the signal to recover the signal envelope. Otherwise, the high-frequency components of the signal will be distorted by low-pass filtering. Therefore, as the sampling frequency of the filter signal increases, the filter signal is not only more fine-grained in frequency but also retains more high-frequency components. From (1), it can be seen that the fine-grained and higher frequency components of $g(x, y)$ result in the preservation of diversity and more high-frequency components for the feature map.

The information from one pixel of infrared images is diffused to the surrounding pixels during the imaging process, which results in blurred edges of objects. The spatial gradient at the edge extracted from the $3 \times 3$ kernel is slight and susceptible to noise disturbance, as shown in the blue box in Fig. 1. Our intuition is that increasing the dilatation of the kernel enables the edge operator to get a more remarkable gradient at the edge due to the broader range it spans. We found experimentally that the kernel with dilation has the same effect on extracting edge features as decreasing its cutoff frequency, which will be further discussed in Section IV-E.

The kaleidoscope module is designed based on the analysis mentioned above and the detailed structure is presented in Fig. 4. In the kaleidoscope, feature maps are first generated in several parts by $1 \times 1$ convolutional layers. Then, involution kernels with large sizes or large dilation rates are applied to extract various features, which are beneficial for infrared object detection and recognition. In addition, global self-attention has an excellent ability for low-frequency feature extraction [43], which allows making full use of the low-frequency features in infrared images.

Therefore, we combine various forms of involution kernels and the self-attention by Inception [44] structure to enrich various frequency features. Finally, feature maps with various frequencies are concatenated and adaptively aggregated by the channel attention [47], which assigns higher weights to critical frequency channels. The weights are obtained by making a global average pooling of feature maps and two linear transformations. The module is named kaleidoscope for two reasons. On the one hand, its kernels are superimposed together like a kaleidoscopic pattern, as shown in Fig. 5. On the other hand, the frequencies of output feature maps are diverse, just like the kaleidoscope.

We employ the involution operator [46] instead of convolution to extract features because the object in the infrared image presents different characteristics at different temperatures, which we also call the problem of multiple patterns. The convolution kernel extracts the fixed features, which will not be updated after the training is finished. When the thermal radiation of the vehicle changes due to the movement state, the local features will also change. The original convolutional kernel may not be good at extracting such features. Unlike convolution, the involution operator is input-adaptive due to its flexible weights, which is related to the first power of the input, as shown in (3). It generates different weights to extract features according to the different patterns of objects, which improves the robustness of the network for multiple patterns. Next, we describe the involution operator in detail, as shown in Fig. 6.

The feature maps are represented as $X \in \mathbb{R}^{C \times H \times W}$. The features at the position of $(i, j)$ are denoted by $X_{ij} \in \mathbb{R}^C$. In the involution, the features $X_{ij} \in \mathbb{R}^C$ are first mapped by a linear transformation $F(\cdot)$ to $A_{ij} \in \mathbb{R}^{K^2}$. $K$ is the size of the kernel. Then, $A_{ij} \in \mathbb{R}^{K^2}$ are reshaped to $A'_{ij} \in \mathbb{R}^{K \times K}$. Specifically, 1-D vectors are unfolded into 2-D kernels. We also add dilation rates to the kernel. Finally, the weights $A'_{ij} \in \mathbb{R}^{K \times K}$ aggregate the features at the corresponding positions $\Delta K$ to yield $Y_{ij} \in \mathbb{R}^{C \times 1 \times 1}$. The equations are given as follows:

$$A_{ij} = F\left(X_{ij}\right) = W_1 \sigma\left(W_0 \cdot X_{ij}\right) \tag{3}$$

$$A_{ij} \xrightarrow{\text{reshape}} A'_{ij} \tag{4}$$

$$Y_{ij} = \sum_{(u,v) \in \Delta K} A'_{i+u,j+v} \cdot X_{i+u,j+v} \tag{5}$$

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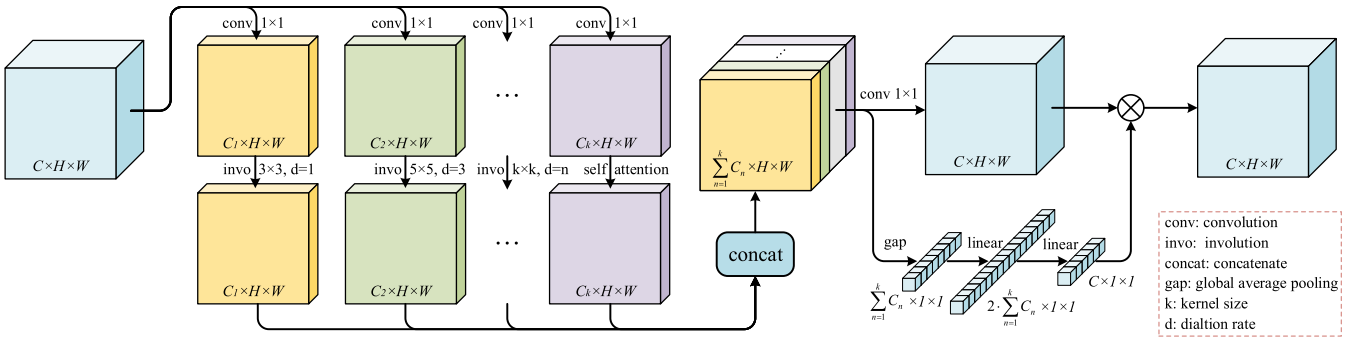



Fig. 4. Schematic of the kaleidoscope module. First, we increase the size and dilation rate of the kernel to increase the cutoff frequency of the low-pass filter and decrease the cutoff frequency of the high-pass filter, respectively. Then, feature maps with various frequencies are concatenated and adaptively aggregated by channel attention.

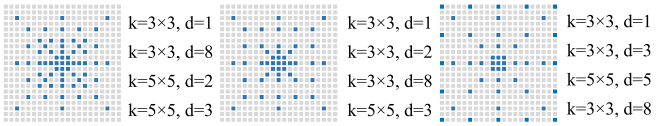



Fig. 5. Diagram illustrating forms of kernels in the kaleidoscope module. They consist of a superposition of various forms of kernels, where $k$ represents the kernel size and $d$ represents the dilation rate.

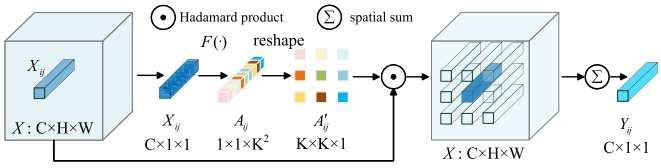



Fig. 6. Illustration of the involution with dilation.

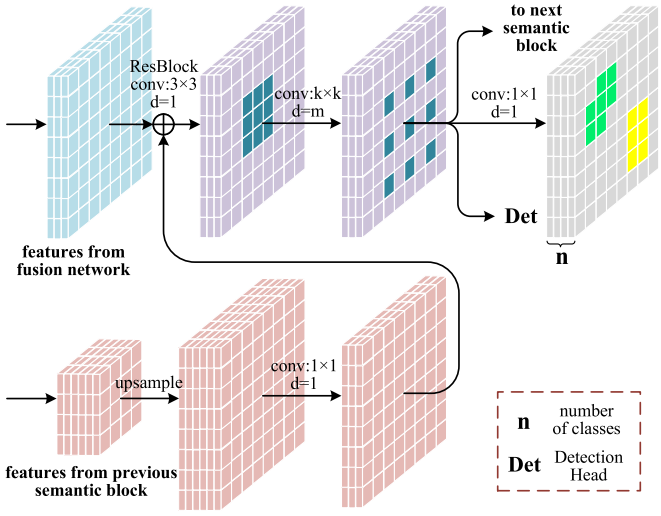



Fig. 7. Illustration of one scale semantic block in the SFSM. It accepts the feature maps from the feature fusion network (the blue part) and the previous scale semantic block (the red part). The purple feature map is the fusion of the two parts. The gray feature map is the one-hot coding for loss calculation.

where $W_0$ and $W_1$ are mapping matrices, $\sigma$ is a batch normalization function, $F$ is a transformation function, and $\Delta K$ is the range of aggregated features.

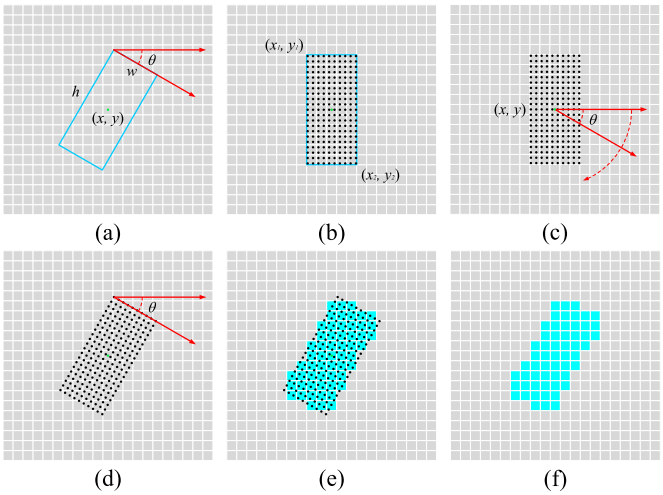



Fig. 8. Diagram of steps to generate the semantic label based on the rotated box annotation. (a) Represents the oriented bounding box of the vehicle. (b)–(d) Depict the region within the oriented bounding box by rotating the horizontal grid. (e) and (f) Display the mask labels obtained by rounding the rotated grid.

### C. Semantic Feature Supplement Module

Due to the shortage of colors and distinct outlines in infrared images, there are insufficient discriminative features for network prediction. Sufficient features are required to predict the position, category, size, and angle of the vehicle in the detection task. For this motivation, we propose the SFSM that is jointly trained with the detection network. The introduction of semantic supervision reduces the optimization difficulty and assists the network in mining hard-to-learn features. Furthermore, the feature map output from the SFSM reflects the information about the location, category, and angle of the vehicle. It is beneficial to improve the detection performance by adding semantic feature maps to the detection head.

The whole SFSM is presented in the green part of Fig. 3, which processes three scales of features by three semantic blocks. We describe one of them in detail, whose structure is shown in Fig. 7. Specifically, the feature map output from the previous scale semantic block is first channel-adjusted by a $1 \times 1$ convolutional layer and upsampled by bilinear interpolation to adjust to the same shape as the current scale feature map. Then, the two parts of the feature map are fused and processed by three residual blocks that consist of

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$3 \times 3$ convolutional layers, a batch normalization, and a rectified linear unit (ReLU) function in sequence. Subsequently, we connect long-range features by a dilation convolution, which has proven beneficial for segmentation tasks [76]. Finally, the extracted feature maps have three uses. They are fused into the next scale semantic block, adjusted to one-hot encoding by the $1 \times 1$ convolutional layer for loss calculation, and supplemented to the detection head. It is worth noting that although some algorithms [25], [73] also fuse semantic segmentation features into the detection, there are differences in the fusion approach compared to our SFSM. Unlike them, we supplement the semantic features directly to the detection features, which strengthens the object features and avoids false suppression of weak objects.

The label for training the SFSM does not require manual annotation. Since the vehicle is almost rectangular in aerial images, we generate the semantic mask directly based on the rotated box, as shown in Fig. 8. The oriented vehicle is labeled as center point $(x, y)$, width $w$, height $h$, and angle $\theta$, as shown in Fig. 8(a). We first ignore the angle and transform the annotation into the form of top-left $(x_1, y_1)$ and bottom-right $(x_2, y_2)$ points to generate the grid points in fixed intervals to cover the box, as shown in 8(b). Then, all grid points are rotated at an angle $\theta$ around the center point of the box $(x, y)$, as shown in Fig. 8(c), to obtain Fig. 8(d). Finally, we round the coordinates of the rotated grid points to the nearest integer, which indicates the semantic mask of the vehicle, as shown in 8(e) and (f).

## IV. Experiments

In this section, we compare our I$^2$MDet with several state-of-the-art approaches on oriented bounding box detection and vehicle detection by quantitative and qualitative comparisons. We first describe the configuration and implementation details of experiments, including the dataset, the evaluation metrics, the network, and the optimizer. Subsequently, we conduct numerous ablation experiments with state-of-the-art alternatives to demonstrate the effectiveness of our designs. Then, we investigate the effect of parameters on the features and results of the network. Finally, we analyze and explain the improvements with visualization.

### A. Experimental Configurations

We verify our approach on the DroneVehicle [63], which is a publicly available aerial image dataset of oriented vehicles. The details of datasets and evaluation metrics are given as follows. DroneVehicle is the largest RGB-Infrared cross-modality vehicle detection dataset on aerial images. It includes 28 439 pairs of infrared and visible images with oriented bounding box annotations of 953 087 instances. Specifically, it is divided into three parts: training, validation, and test sets, with 17 990, 1469, and 8980 images, respectively. The scale of images is $840 \times 712$ pixels, which extends 100 pixels of a white border in the original $640 \times 512$ size. The dataset annotates five categories: car, truck, bus, van, and freight car. In addition, the DroneVehicle covers multiple scenarios, such as urban roads, residential areas, and parking lots from day to night. In this article, we only adopt infrared images of the dataset and cut them into a square of $704 \times 704$ for operational purposes. Since the removed part is the white border, it does not affect the experimental results.

We selected three indicators to quantify the approach, including the mean average precision (mAP), the number of parameters (Params), and the floating-point operations (FLOPs). The mAP is to evaluate the effectiveness of the model. We follow the mAP@0.5 and mAP@.5:.95 of COCO protocol. The Params and FLOPs are used to measure the capacity of the model for a more comprehensive evaluation.

### B. Implementation Details

We set four types of involution operators in the kaleidoscope module. Their kernel sizes and dilation rates are 3, 3, 5, and 5 and 1, 8, 2, and 3, respectively. The kernel sizes and dilation rates of three dilation convolutions in the SFSM are 5, 5, and 7 and 2, 4, and 4, respectively. It has been experimentally proven that the network outputs the best result with these settings. We will discuss in detail how the kernel parameters influence the network features in Section IV-E, which guides us in the selection of parameters. The batch size is set to 32 in the dataloader. We perform augmentation to images in the training stage, such as flip, crop, and mosaic. The parameters of our I$^2$MDet are optimized by stochastic gradient descent (SGD) with momentum. The momentum and the weights decayed are set to 0.93 and $5 \times 10^{-4}$, respectively. The learning rate is initially set to $5 \times 10^{-3}$ and then decayed exponentially from 0 to 400 epochs. The comparison experiments in this article are implemented in the MMRotate platform [78]. All experiments are conducted on the PyTorch framework [79] and NVIDIA GeForce RTX 3090.

### C. Comparison to State of the Art

In this section, we compare our I$^2$MDet with several state-of-the-art methods on oriented bounding box detection and vehicle detection described in related work on DroneVehicle. Most oriented object detection methods we compare are proposed for detecting dense and arbitrarily oriented vehicles in aerial images.

*1) Results on DroneVehicle:* We show the results of comparisons between I$^2$MDet and state-of-the-art algorithms in Table I. It can be seen that our I$^2$MDet achieves the best performance on mAP@.5 and mAP@.5:.95. It also obtains the highest score on five vehicle subcategories. Furthermore, our approach is very competitive on the number of parameters and the FLOPs, which has the smallest FLOPs and the second smallest Params. Most of the approaches employed in our comparison experiments are ranked at the top of the DOTA leaderboard. Compared to RoI Transformer and KFIoU, which achieve the best performance on the DOTA dataset until July 2022, the I$^2$MDet improves by 7.01/9.30 and 2.63/5.98 on mAP@.5 and mAP@.5:.95, respectively. UA-CMDet is designed based on infrared-visible fusion for DroneVehicle. Our method increases by 13.30 on mAP@.5 in the case of exploiting infrared images only compared to UA-CMDet.

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TABLE I

RESULTS OF THE COMPARISON EXPERIMENT ON DRONEVEHICLE. THE EVALUATION METRIC FOR THE VEHICLE SUBCATEGORY IS AP@.5. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS OF THE MAP. PARAMS AND FLOPS ARE THE NUMBER OF PARAMETERS AND THE FLOATING-POINT OPERATIONS, RESPECTIVELY, TO MEASURE THE CAPACITY OF THE MODEL. **RED** INDICATES THE BEST RESULT. **BLUE** INDICATES THE SECOND BEST RESULT

| Methods | car | truck | van | bus | freight | mAP@.5 | mAP@.5:.95 | Params.(M) | FLOPs(G) |
|---|---|---|---|---|---|---|---|---|---|
| Rotated RetinaNet [17] | 88.3 | 20.7 | 19.6 | 66.2 | 32.5 | 45.47 | 27.76 | 36.21 | 99.95 |
| Rotated Faster R-CNN [15] | 89.9 | 48.9 | 45.8 | 86.9 | 48.3 | 63.67 | 36.04 | 41.13 | 107.20 |
| Rotated RepPoints [77] | 89.4 | 50.4 | 42.9 | 72.2 | 42.1 | 59.39 | 35.82 | 36.61 | 91.87 |
| Rotated FCOS [18] | 89.8 | 48.6 | 43.4 | 88.5 | 50.2 | 64.08 | 40.07 | 31.90 | 97.59 |
| Rotated ATSS [19] | 89.6 | 38.2 | 26.5 | 84.9 | 28.2 | 53.48 | 32.52 | 36.02 | 98.04 |
| RoI Transformer [23] | 90.1 | 60.4 | **52.2** | 89.7 | **58.9** | **70.29** | **43.57** | 55.06 | 121.13 |
| Gliding Vertex [28] | 89.9 | 47.0 | 44.4 | 88.1 | 49.1 | 62.96 | 35.04 | 41.13 | 107.20 |
| CSL [27] | 88.4 | 22.4 | 22.5 | 60.0 | 32.1 | 44.96 | 26.77 | 37.14 | 109.58 |
| $R^3$Det [26] | 89.8 | 48.2 | 44.5 | 88.1 | 49.6 | 63.75 | 36.77 | 36.95 | 107.63 |
| UA-CMDet [63] | 87.5 | **60.7** | 38.0 | 87.1 | 46.8 | 64.00 | - | - | - |
| $S^2$ANet [30] | 89.8 | 45.2 | 39.4 | 88.8 | 44.6 | 61.52 | 35.30 | 38.55 | 92.96 |
| ReDet [31] | **90.2** | 57.6 | 48.3 | 89.7 | 54.5 | 68.09 | 42.44 | **31.57** | **52.56** |
| CFA [32] | 90.0 | 58.7 | 48.0 | 88.5 | 51.7 | 67.31 | 42.59 | 36.60 | 91.85 |
| Oriented R-CNN [33] | 89.9 | 56.6 | 46.9 | 89.6 | 54.4 | 67.52 | 42.60 | 41.13 | 107.26 |
| GWD [36] | 86.8 | 24.8 | 13.8 | 72.3 | 28.0 | 45.04 | 26.02 | 36.21 | 99.95 |
| KLD [37] | 89.9 | 39.7 | 33.7 | 87.0 | 35.5 | 57.17 | 34.58 | 41.67 | 156.37 |
| SASM [35] | 89.7 | 51.9 | 38.0 | 86.2 | 49.8 | 62.88 | 40.65 | 36.60 | 91.85 |
| Oriented RepPoints [34] | 89.9 | 54.8 | 47.5 | 88.4 | 52.2 | 66.98 | 41.32 | 36.60 | 91.85 |
| KFIoU [38] | 90.1 | 56.2 | 51.7 | **89.8** | 52.3 | 68.00 | 40.22 | 55.06 | 121.13 |
| PSC [56] | 89.9 | 52.2 | 47.9 | 88.9 | 53.4 | 66.40 | 41.71 | 31.91 | 97.71 |
| ViT-B+RVSA [57] | 89.7 | 52.3 | 44.4 | 88.0 | 51.0 | 65.07 | 42.63 | 60.46 | 134.69 |
| $I^2$MDet(ours) | **96.3** | **73.4** | **58.6** | **93.2** | **65.0** | **77.30** | **46.20** | **31.87** | **48.90** |

TABLE II

ABLATION STUDY OF KALEIDOSCOPE AND SFSM ON DRONEVEHICLE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS. BOLD INDICATES THE BEST RESULT

| Method | Modules | | mAP@.5 | mAP@.5:.95 |
|---|---|---|---|---|
| | kaleidoscope | SFSM | | |
| $I^2$MDet | × | × | 74.12 | 42.85 |
| | CoordAtten [80] | × | 75.44 | 45.34 |
| | CBAM [81] | × | 75.67 | 44.64 |
| | PPM [82] | × | 76.12 | 44.88 |
| | convolution | × | 75.35 | 43.76 |
| | ✓ | × | 76.41 | 45.47 |
| | × | ✓ | 76.27 | 45.19 |
| | ✓ | InLD [25] | 76.81 | 45.86 |
| | ✓ | ✓ | **77.30** | **46.20** |

*2) Output Visualization:* According to the results of experiments in Table I, we select five methods with superior performance and representation for output visualization. They are RoI Transformer [23], KFIoU [38], ReDet [31], Rotated Faster R-CNN [15], and Oriented RepPoints [34]. RoI Transformer with KFIoU gets the highest score on the DOTA dataset until July 2022. ReDet achieves competitive performance with the smallest Params and the second smallest FLOPs. Rotated Faster R-CNN, one of the most famous methods, is used as the baseline for improvement by many approaches. Unlike previous anchor-based approaches, the object in Oriented RepPoints is represented by a set of points, which allows more flexibility in extracting features of the object. The output visualization of the five advanced methods and our $I^2$MDet is shown in Fig. 9. Due to the absence of color information in the infrared image, many objects are similar to vehicles. This leads detection methods designed for visible images to generate many false alarms on infrared images. In Fig. 9(a), except for our $I^2$MDet, the other methods recognize the bus station beside the road as a bus. The roof in Fig. 9(b) and the container in Fig. 9(d) and (e) are mistakenly detected as freight cars. Cabs of excavators are detected as cars in Fig. 9(b) and (d). The SFSM in $I^2$MDet supplements the detection head with semantic features containing category information, which improves the performance of distinguishing similar objects in infrared images. The patterns of cars in the infrared image are quite diverse, as shown in Fig. 9(f), leading to the false alarms and the undetected ones which are most likely to generate in the category of cars. In addition, some vehicles have blurred edges and textures in infrared images, presenting their ambiguous appearances. The kaleidoscope module can focus on a wider frequency range of features and alleviate the problem of multiple patterns. Therefore, $I^2$MDet can recognize vehicles inconspicuous in the background, such as the car in the bottom-right corner of Fig. 9(f).

### D. Ablation Study

To verify the effectiveness of the kaleidoscope and the SFSM in $I^2$MDet, we conduct some ablation studies on these two improvements. The experimental results are shown in Table II. The baseline is our $I^2$MDet without kaleidoscope modules and semantic supervision, which is shown in the first row of Table II.

*1) Ablation on Kaleidoscope:* We first demonstrate the improvement of the kaleidoscope module through two groups of comparison experiments. See the first and sixth rows in Table II, which shows that the baseline is increased by 2.29 on mAP@.5 and 2.62 on mAP@.5:.95 by kaleidoscope modules. As indicated in rows 7 and 9 of Table II, kaleidoscope modules increase the baseline with the SFSM by 1.03 on

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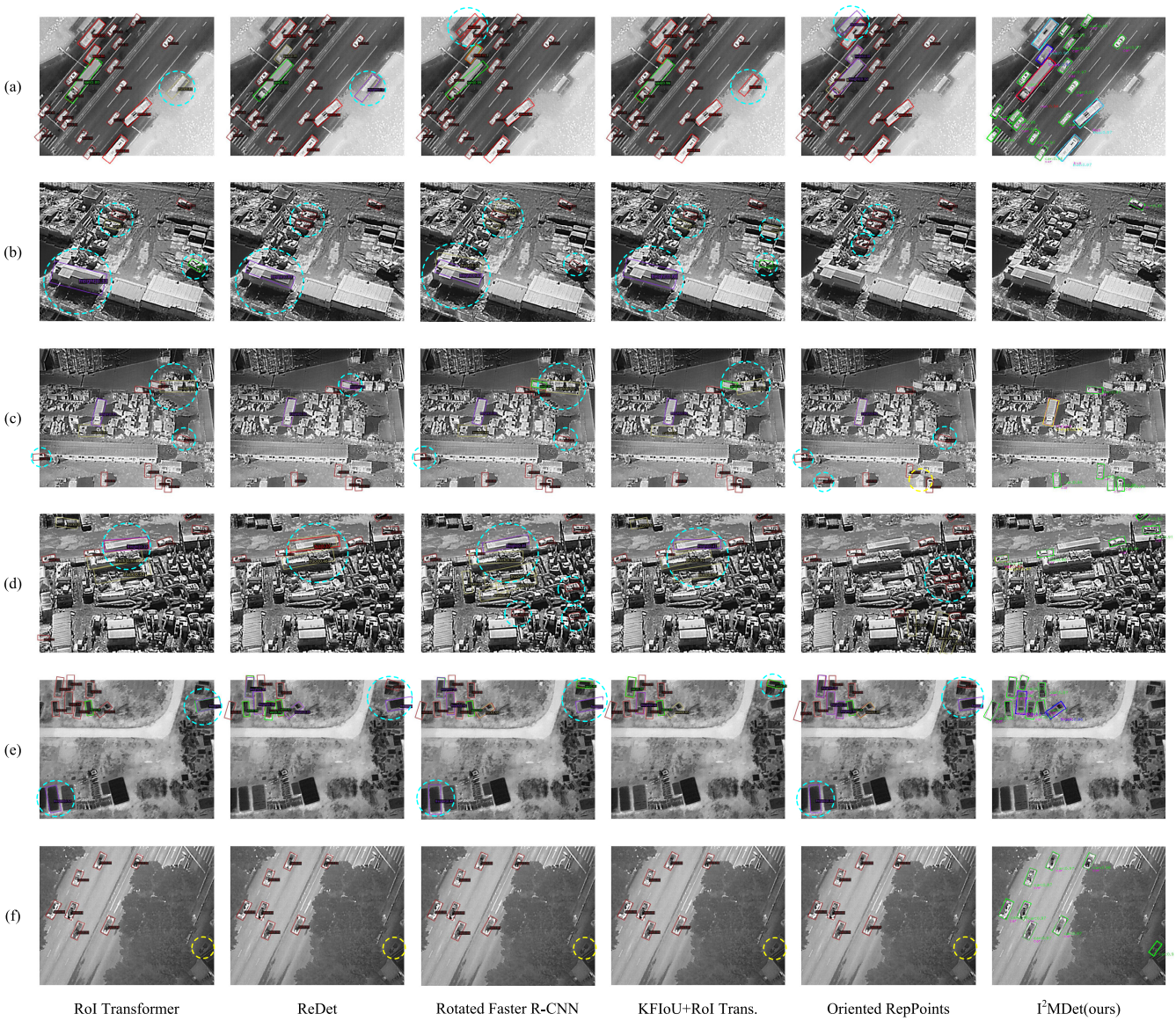



Fig. 9. Illustration of output visualization. (a)–(f) Samples in the test dataset. The approaches we select are RoI Transformer, ReDet, Rotated Faster R-CNN, KFIoU, Oriented RepPoints, and our I$^2$MDet. The cyan and yellow dashed circles in the image indicate that the object has been mistakenly detected and not detected, respectively. The ground-truth boxes and categories are marked in pink on the images in the last column.

mAP@.5 and 1.01 on mAP@.5:.95. Then, to prove that the kaleidoscope module performs better on infrared images than other enhanced approaches, we replace the kaleidoscope in our framework with two advanced attentions: coordinate attention (CoordAtten) [80] and convolutional block attention module (CBAM) [81]. The experimental results are shown in rows 2 and 3 of Table II, where our kaleidoscope outperforms CoordAtten and CBAM. A pyramid pooling module (PPM) [82] fuses various frequency features by multiscale pooling, which is similar to the purpose of our kaleidoscope. The fourth row of experiments demonstrates that PPM is inferior to our module. We will analyze the difference between PPM and the kaleidoscope module in Section IV-E. In the fifth experiment of Table II, we replace the involution in the kaleidoscope with the convolution. The performance shows that the involution operator is more beneficial to extracting features in infrared images. This is because the patterns of vehicles in infrared images are complex and diverse. It is difficult to extract some vehicle features with fixed convolution filters. In contrast, the weights of the involution filter are related to the first power of the input, so it is more robust to complex vehicle features in infrared images.

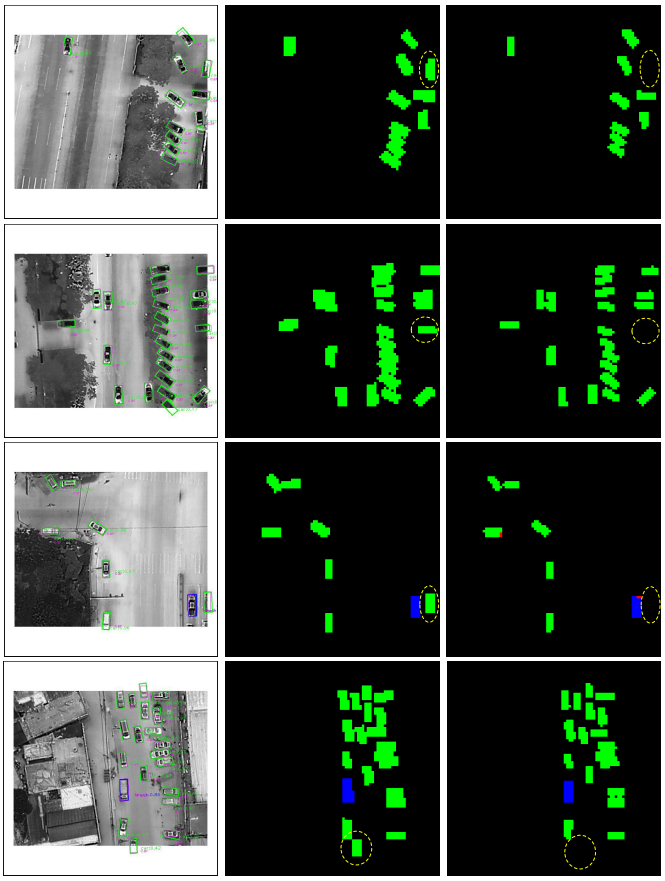
TABLE III

EFFECT OF THE KERNEL SIZE IS STUDIED ON DRONEVEHICLE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS. BOLD INDICATES THE BEST RESULTS

| kernel sizes | dilation rates | mAP@.5 | mAP@.5:.95 |
|---|---|---|---|
| 3 × 3 | 1 | 74.63 | 42.75 |
| 5 × 5 | 1 | 74.90 | 43.31 |
| 7 × 7 | 1 | **74.93** | **43.52** |
| 9 × 9 | 1 | 74.65 | 42.92 |

*2) Ablation on SFSM:* Two groups of experiments illustrate the improvement from the SFSM. The baselines are improved

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(a) (b) (c)

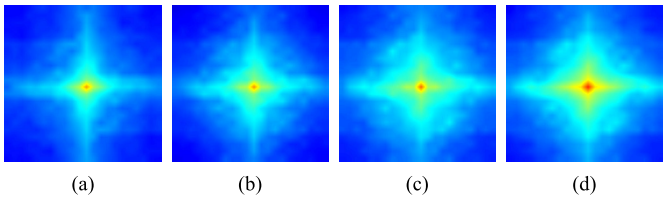
Fig. 10. Output visualization demonstrating the impact of InLD. (a) Detection results without InLD. (b) and (c) Ground-truth masks and denoising masks predicted by InLD, respectively. The vehicles in the yellow dashed circles will be suppressed by the denoising masks.

TABLE IV

EFFECT OF THE DILATION RATE OF THE KERNEL IS STUDIED ON DRONEVEHICLE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS. BOLD INDICATES THE BEST RESULT

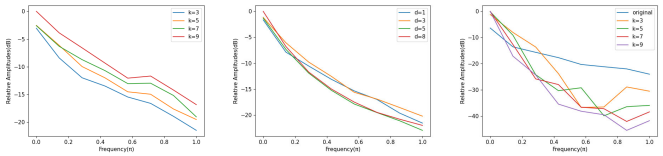
| kernel sizes | dilation rates | mAP@.5 | mAP@.5:.95 |
|---|---|---|---|
| 3 × 3 | 1 | 74.63 | 42.75 |
| 3 × 3 | 2 | 74.68 | 42.56 |
| 3 × 3 | 3 | 74.85 | 43.07 |
| 3 × 3 | 5 | 74.92 | 43.14 |
| 3 × 3 | 8 | **75.01** | **43.39** |
| 3 × 3 | 13 | 74.75 | 42.83 |
| 5 × 5 | 1 | 74.90 | 43.31 |
| 5 × 5 | 2 | 75.17 | 43.35 |
| 5 × 5 | 3 | **75.26** | **43.44** |
| 5 × 5 | 5 | 75.13 | 43.20 |
| 5 × 5 | 8 | 75.10 | 42.87 |
| 5 × 5 | 13 | 75.07 | 42.94 |
| 7 × 7 | 1 | **74.93** | **43.52** |
| 7 × 7 | 3 | 74.87 | 43.38 |
| 7 × 7 | 5 | 74.86 | 43.33 |
| 7 × 7 | 8 | 74.90 | 43.15 |
| 7 × 7 | 13 | 74.74 | 42.80 |

by 2.15 and 2.34 on mAP@.5 and mAP@.5:.95 via applying semantic supervision and complementing the detection head with semantic features comparing the 1st with 7th rows in Table II. On the baseline with the kaleidoscope, the SFSM allows the model to enhance by 0.89 and 0.73 on mAP@.5 and mAP@.5:.95, respectively. The results are shown in the sixth and ninth experiments in Table II. It is worth noticing that the InLD in SCRDet++ [25] is similar to our SFSM but essentially different in its operation. First, it assigns dedicated channels to the features of each category and predicts a mask for it. Then, the mask is applied to the feature map to suppress noise in the background. In contrast, we add the feature maps extracted from the SFSM to the features of the detection head to complement rather than filter the information. In the eighth row of Table II, we apply the InLD in our $I^2$MDet. The performance of the InLD is inferior to our SFSM. Because the predicted mask in InLD is not precise, it can suppress background noise but may also eliminate object features. As shown in Fig. 10, we visualize the detection result of the network and the denoising mask predicted by InLD. Although the denoising mask is very close to the ground truth, there are still individual vehicles missed, as shown by the yellow dashed circle in Fig. 10(b) and (c). Fig. 10(a) shows that the network without InLD can detect all vehicles. If the denoising mask predicted in Fig. 10(c) is adopted to conduct InLD, it will result in the elimination of the vehicles originally detected by the detector.



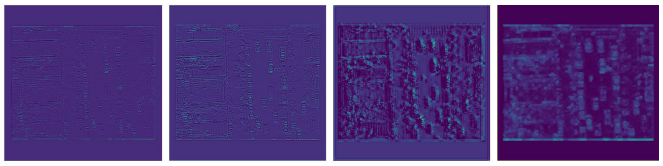
(a) (b) (c) (d)

Fig. 11. Spectrogram of the feature maps. (a)–(d) Spectrograms of the feature maps extracted from the kernels whose sizes are 3 × 3, 5 × 5, 7 × 7, and 9 × 9, respectively.



(a) (b) (c)

Fig. 12. Quantification of spectrograms. (a)–(c) Quantitative spectrograms about kernel sizes, dilation rates, and pooling sizes, respectively.



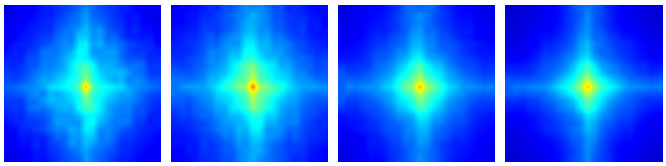
(a) (b) (c) (d)

Fig. 13. Visualization of extracted features from the same size kernels with different dilatation rates. (a)–(c) Feature map extracted from the 3 × 3 kernel with a dilation rate of 1, 3, and 8, respectively. (d) Result of (a) after 7 × 7 max pooling.

### E. Parameter Analysis

In this section, we explore the effect of the kernel size and the dilation rate on the frequencies of extracted features through experiments and visualizations, which guide us to

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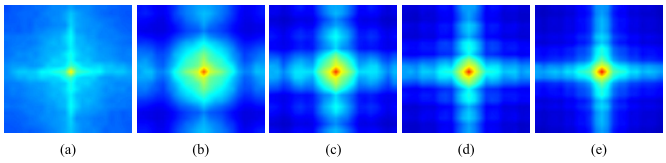
(a) (b) (c) (d)

Fig. 14. Spectrogram of the feature maps. (a)–(d) Spectrograms of the feature maps extracted from the 3 × 3 kernels with one, three, five, and eight dilation rates, respectively.

TABLE V

EFFECT OF PARAMETERS IN THE KALEIDOSCOPE MODULE IS STUDIED ON DRONEVEHICLE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS. BOLD INDICATES THE BEST RESULT

| kaleidoscope | | mAP@.5 | mAP@.5:.95 | Params.(M) | FLOPs(G) |
|---|---|---|---|---|---|
| kernel sizes | dilation rates | | | | |
| 3, 3, 3, 3 | 1, 3, 5, 8 | 75.77 | 44.82 | 16.11 | 27.8 |
| 3, 5, 7, 9 | 1, 1, 1, 1 | 75.98 | 45.14 | 26.61 | 41.4 |
| 3, 3, 3, 5 | 1, 2, 8, 3 | 76.25 | 45.19 | 17.42 | 29.5 |
| 3, 3, 5, 5 | 1, 8, 2, 3 | **76.41** | **45.47** | 18.73 | 31.2 |
| 3, 3, 5, 5 | 2, 8, 2, 3 | 76.13 | 45.22 | 18.73 | 31.2 |



(a) (b) (c) (d) (e)

Fig. 15. Spectrogram of the feature maps. (a)–(e) Spectrograms of the feature maps are max-pooled by 1 × 1, 3 × 3, 5 × 5, 7 × 7, and 9 × 9, respectively.

design the kaleidoscope module. In addition, we discuss the differences between maximum pooling and our approach in terms of achieving feature frequency diversity.

*1) Parameter Analysis on Kernel Sizes:* The relationship between the kernel size on feature frequency and performance is investigated. We replace the kaleidoscope module after CSPBlock 5 with an involution operator and change the kernel size to show the performance in Table III. The performance is improved and then deteriorated with the increase of the kernel size, peaking at 7. Then, we analyze the frequency of feature maps extracted by different size kernels. Fig. 11 shows the spectrograms of features extracted by four-size kernels (3 × 3, 5 × 5, 7 × 7, and 9 × 9), and the quantitative result is presented in Fig. 12(a). They illustrate that the high-frequency component of features will increase as the kernel size increases, which is the same as our evidence in Section III-B. We attribute the performance degradation at k = 9 to more noise than meaningful information passing through the low-pass filter when the cutoff frequency is excessive. Many studies [83], [84] have demonstrated that large kernels provide better results, which we have rethought and explained in the frequency perspective.

*2) Parameter Analysis on Dilation Rates:* To prove that increasing the dilation rates of kernels can decrease the cutoff frequency of high-pass filters, we visualize the feature map extracted from the same 3 × 3 kernel with different dilation rates, as shown in Fig. 13. From Fig. 13(a) and (b), the kernel with dilation extracts more distinct edges, which is similar to the effect of the high-pass filter with low cutoff frequency in Fig. 1. The spectrogram Fig. 14 and its quantization Fig. 12(b) also demonstrate that the features extracted from the kernel with dilation have more low-frequency components. Fig. 13(c) and (d) shows the feature maps extracted by a 3 × 3 kernel with eight dilation rates and a 7 × 7 max pooling, respectively, which show a similar result. We believe that the appropriate dilation rate enhances the features accurately, but excessive dilation causes features to be enhanced repeatedly, which, like max pooling, can blur the location of features. We explore kernel parameters experimentally to find the optimal dilation rate to enhance the features, which is shown in Table IV. We can make two conclusions from Table IV: 1) the large dilation rate does not constantly improve the detection performance and 2) the larger the kernel size is, the smaller the dilation rate should be.

TABLE VI

EFFECT OF THE POOLING SIZE IN PPM IS STUDIED ON DRONEVEHICLE. WE ADOPT MAP@.5 AND MAP@.5:.95 IN THE COCO PROTOCOL AS EVALUATION METRICS

| pooling size | mAP@.5 | mAP@.5:.95 |
|---|---|---|
| 3 × 3 | 74.58 | 43.25 |
| 5 × 5 | 75.65 | 43.57 |
| 7 × 7 | 75.72 | 43.64 |
| 9 × 9 | 75.79 | 43.89 |
| 3 × 3 + res | 75.19 | 44.07 |
| 5 × 5 + res | 75.82 | 44.94 |
| 7 × 7 + res | 76.10 | 44.85 |
| 9 × 9 + res | 76.08 | 44.90 |
| 3, 5, 7, 9 + res | 76.12 | 44.88 |

*3) Parameter Analysis on the Kaleidoscope Module:* In this section, we discuss whether combining features of diverse frequencies can further improve detection performance. Following the regularity explored in experiments of kernel sizes and dilation rates, we design several parameter combinations of the kaleidoscope module, as shown in Table V. The first and second rows of experiments demonstrate that diverse features are beneficial to the improvement of the result. Then, we select excellent kernels in Tables III and IV to form the kaleidoscope module, which is shown in the third, fourth, and fifth rows of Table V. The fourth and fifth rows of experiments indicate that the kernel of d = 1 is indispensable, which has a role as the residual structure to preserve the fundamental frequency components. Thus, the kaleidoscope module in our network selects the parameters of the fourth experiment.

*4) Effect of Pooling on Feature Frequency:* The kaleidoscope module enhances the detection performance by improving low-frequency features and high-frequency features and fusing them, which can also be interpreted as increasing the diversity of feature frequency. Some other works also enhanced the frequency diversity to improve the accuracy. For example, iFormer [43] broadens the frequency range of ViT [85] by adding max pooling. PPM in PSPNet [82] allows the network to focus on features in different regions through multiscale pooling, which is also a way to increase frequency diversity. We add different sizes of max pooling

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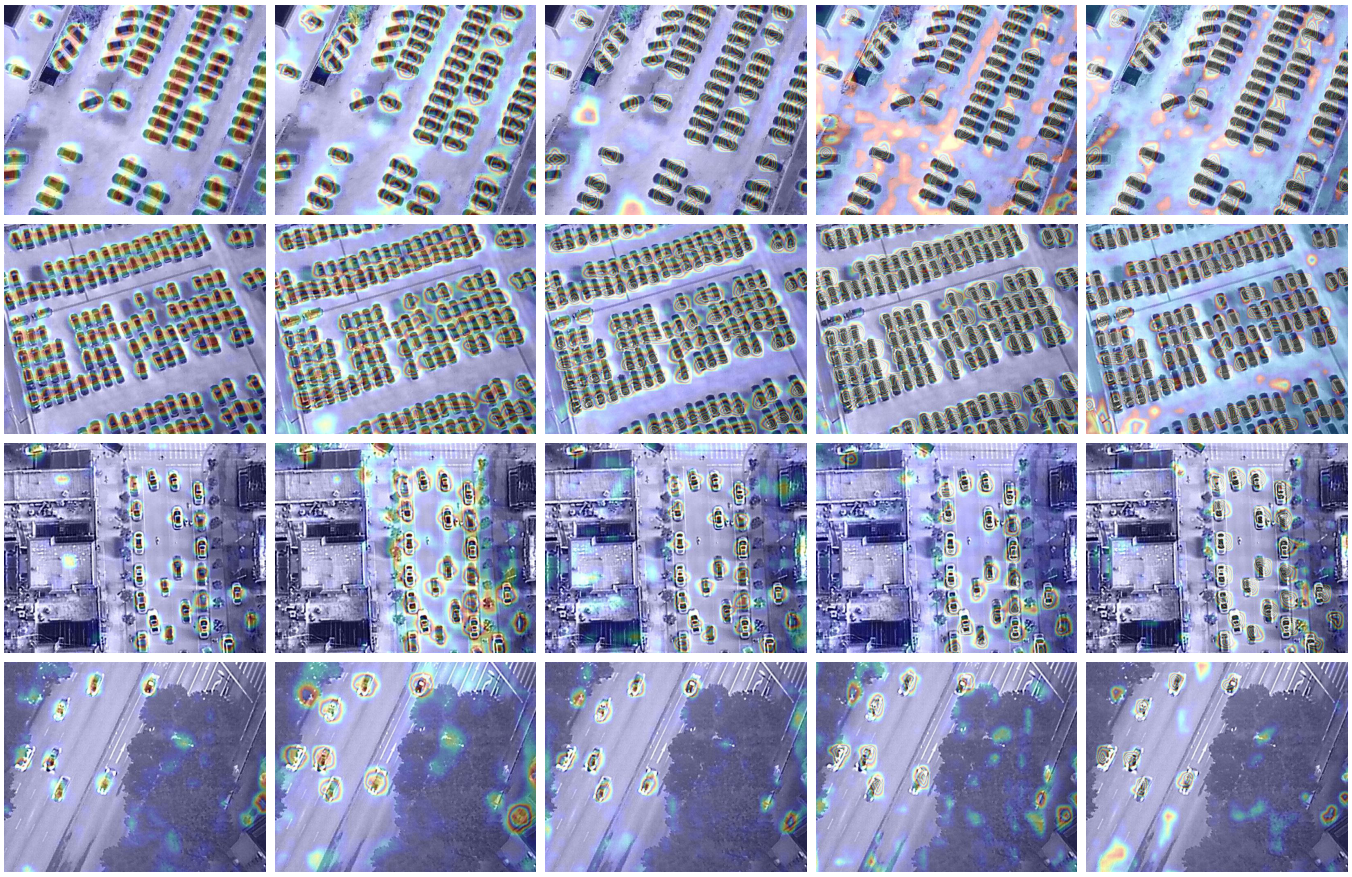



Fig. 16. Visualization of the feature maps from the kaleidoscope module. (a)–(e) Frequency of features gradually increases.

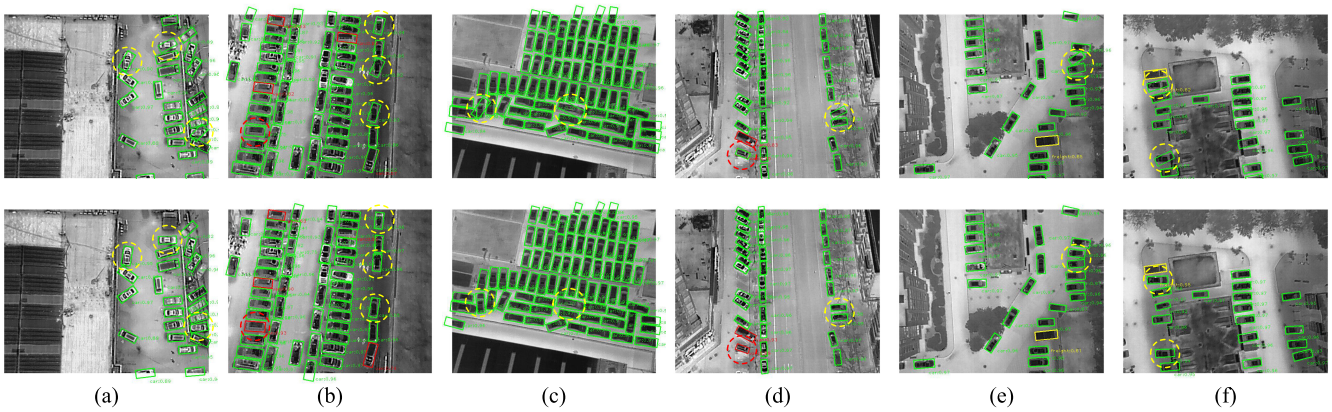



Fig. 17. Visualization of the role of the SFSM. (a)–(f) Samples in the test dataset. The first and second rows are the detection results without and with the SFSM, respectively. The yellow dashed circle indicates the object whose angle is corrected. The red dashed circle indicates the object whose category is corrected.

after CSPBlock 5 and the experimental results are shown in Table VI. From the first part of Table VI, it can be seen that the performance improvement trend becomes slower as the pooling area increases. The second part of Table VI shows the existence of performance saturation for frequency diversity achieved by multiscale pooling. Furthermore, we analyze this phenomenon from the perspective of frequency. The spectrograms of features pooled at different scales are provided in Fig. 15 and their quantitative results are presented in Fig. 12(c), which show that as the max-pooling area widens, the low-frequency component of the feature increases and the high-frequency component decreases. From the perspective of signal sampling, max pooling only samples the maximum value in a specific area and discards the minimum value, which causes the signal envelope to become smooth. In other words, part of the high-frequency component is lost. Although it can enhance the low-frequency global features of the object and suppress high-frequency noise, it does not extend the

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frequency range and provide a fine-grained frequency component. This is the reason for the performance saturation of frequency diversity achieved by multiscale pooling. Unlike pooling, expanding the kernel size provides a fine-grained frequency component and widens the cutoff frequency, which has a higher performance ceiling.

### F. Visualization and Discussion

For a more intuitive understanding of the role played by the kaleidoscope and the SFSM in the network, we perform some visualizations in this section.

The activation region in the feature map can reflect the features that the network focuses on. Overlaying the feature map on the original image can help us to understand the attention of the network. We visualize the feature map of the kaleidoscope module on the original image and find that the activated features of vehicles show various frequency distributions, as shown in Fig. 16, which demonstrates that the kaleidoscope module can adequately capture the vehicle characteristics in various frequencies. Fig. 9(f) and the last row of Fig. 16 are the same image. How our I$^2$MDet detects the vehicle in the bottom-right corner of Fig. 9(f) can be interpreted as the superposition of multiple-frequency features together contributing to the recognition.

The detection results with and without the SFSM are presented in Fig. 17 to explore the qualitative and quantitative impact of the SFSM on the network. The first row of Fig. 17 shows the result of I$^2$MDet after removing the SFSM, and the second row shows the development of our I$^2$MDet with the SFSM. The vehicle in the red dashed circle of Fig. 17(b) and (d) is detected as the correct category after adding the SFSM because the edges of the vehicle are blurred in the infrared image. When the van is arranged densely among cars in Fig. 17(b), the feature extraction of the van will be disturbed by the surrounding cars. The SFSM, on the one hand, increases the number of positive samples and introduces definite marginals for supervision, and on the other hand, complements the detection head with semantic features containing category and position information. They lead to more refined vehicle features in the detector. The yellow dashed circle in Fig. 17 shows that the introduction of the SFSM promotes the angle regression of the network due to the contribution of angular information in semantic features.

## V. Conclusion

In this article, we propose a novel framework for oriented vehicle detection of infrared aerial images called I$^2$MDet. We design a kaleidoscope module and an SFSM to tackle three challenges: blurred edges, multiple patterns of objects, and little available information in infrared images. The kaleidoscope adopts kernels with various sizes and dilation rates to focus on features at different frequencies. We employ the involution operator to extract features in the kaleidoscope to enhance robustness toward multiple patterns because of the input self-adaptability of involution. We devise an SFSM and adopt a supplemental fusion to solve the problem of insufficient information in infrared images. On the one hand, semantic supervision guides the network to learn more meaningful features. On the other hand, the features containing information about the category, location, outline, and angle of the object are added to the detection head, which contributes to classification and regression. We conduct extensive comparison experiments and ablation studies to demonstrate the superiority of our designs. Through theorems and experiments, we explore the relationship between kernel parameters and feature frequency both qualitatively and quantitatively, providing theoretical support for the design of our kaleidoscope module. Furthermore, we analyze the role of pooling in the network from a frequency perspective. Finally, we visualize the detection results and feature maps of the network to evaluate the effectiveness of our designs.

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